IncidentLens: Inferring City-Scale Incidents from Heterogeneous Urban Observations

Brian Wang
University of California, Los Angeles
Los Angeles, California, USA
wangbri1@g.ucla.edu

Oliver Wang
University of California, Los Angeles
Los Angeles, California, USA
owang22@g.ucla.edu

Gaofeng Dong
University of California, Los Angeles
Los Angeles, California, USA
gfdong@g.ucla.edu

Mani Srivastava
mbs@ucla.edu
University of California, Los Angeles
Los Angeles, California, USA

Abstract

Urban emergencies and disasters can produce widespread consequences that are difficult to recognize from any single observation stream. The Palisades Fire and the 2008 Mumbai attacks illustrate two different forms of this problem: one evolved through physical processes across a wildland-urban interface, while the other unfolded through coordinated human actions across multiple urban sites. We describe such unplanned emergencies and disasters that extend across large regions of space and time as *incidents*.

Detecting incidents is important for warning, evacuation, response allocation, situational awareness, and resource coordination. However, city-scale incident detection differs from conventional anomaly detection or local event classification because available sensing streams often observe only indirect, partial, and unevenly distributed effects. As a result, the system must infer latent incident properties, including type, origin, extent, evolution, and complex spatiotemporal relationships among local observations, from heterogeneous evidence.

This paper presents *IncidentLens*, a system for detecting incidents from heterogeneous urban observation streams. *IncidentLens* converts raw sensor and text inputs into normalized semantic observations and maintains competing incident explanations over space and time. As new observations arrive, the system updates, prunes, and clusters these explanations to produce incident detections that link distributed evidence into coherent event-level records.

We evaluate *IncidentLens* on controlled synthetic incidents and real Los Angeles incident data. On synthetic low-level incidents, *IncidentLens* achieves 0.664 F2, 0.762 recall, and 0.98 false positives per run, improving F2 by 3.9x over the strongest baseline while substantially reducing false positives. On real incidents, *IncidentLens* achieves the best F2 among evaluated methods, matching 17 of 29 reported incidents with 79 false positives and a median detection delay of -29.2 hours relative to the earliest external article timestamp. These results suggest that our approach can improve weak-label event detection under sparse, indirect, and multimodal urban observability, while also exposing remaining challenges in false-positive control and spatial localization on real data.

CCS Concepts

• **Information systems → Spatial-temporal systems**; *Sensor networks*; • **Computing methodologies → Spatial and physical reasoning**.

Keywords

urban sensing, incident detection, multimodal sensing, spatiotemporal reasoning, large language models

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1 Introduction

Emergencies and disasters occur in cities throughout the world, posing threats to human life, property, critical infrastructure, and public safety. These emergencies and disasters can unfold over extended periods of time, affect broad geographic regions, and generate various observable effects.

We illustrate some of the interesting behaviors of these emergencies and disasters in Fig 1. The 2025 Palisades Fire began in the Santa Monica Mountains near Pacific Palisades and evolved over days into a large wildland-urban interface fire, driven by wind, terrain, fuel, and suppression dynamics. In addition, this wildfire's observable footprint was not limited to just fire: smoke, air quality changes, evacuations, road congestion, structure damage, and public reports appeared across a much broader region. In contrast, the 2008 Mumbai attacks unfolded through coordinated human actions at multiple urban sites, including transportation, hospitality, medical, and community locations. Rather than spreading continuously through physical fuel and wind, the Mumbai attacks evolved through discrete, coordinated actions and emergency responses. Both of these examples are instances of *incidents*: unplanned emergencies or disasters whose effects may extend across space and time. These examples also show that some incidents are compositional, where multiple local fires, attacks, closures, evacuations, or response activities must be interpreted as related parts of a larger whole.

Timely detection of incidents is critical because early recognition can support warning, response and resource allocation, evacuation, situational awareness, and multi-agency coordination. Fortunately, government agencies, companies, and citizens provide heterogeneous sensing streams—traffic measurements, air-quality monitors, public cameras, emergency reports, social media, and news—that could reveal incidents earlier than any single modality. However, these streams rarely observe the full incident directly. A camera may see haze rather than a fire front, a traffic sensor may see congestion rather than an evacuation, an air-quality sensor may see particulate matter rather than an ignition, and public reports may be delayed or spatially coarse. The problem of detecting incidents is therefore not only to detect local anomalies, but to infer the incident that explains why distributed observations are related.

Existing urban sensing systems typically address a different abstraction. Anomaly detection methods identify deviations at sensors, road segments, regions, or social clusters, but these methods often do not explain the relationships among observations distributed across larger spatial regions or longer time periods. Event classifiers can recognize specific local phenomena, such as crashes, smoke, floods, or crowds, but are usually specialized to a particular modality or incident type. Source localization and inverse-modeling methods reason backward from effects to causes, but often assume that the incident type, propagation model, and relevant observations are known in advance. City-scale incident detection violates these assumptions: incident types are diverse, severe incidents are rare, labels are weak or delayed, sensors are unevenly deployed, and relevant evidence may appear far from the source that caused it.

This paper studies the problem of detecting incidents from physical, mobility, environmental, visual, and public-report evidence in available sensing streams. The key difficulty is that the incident itself is only partially observed. Its type, origin, spatial extent, temporal evolution, and relationships to other local activities may be unknown. We therefore treat these properties as *latent*: they must be inferred from indirect and heterogeneous effects rather than read directly from any single sensor.

This setting creates three core challenges. First, observations are *indirect*: sensors often see effects of an incident rather than the incident source itself. Second, incidents obey *different physics*. A wildfire evolves continuously through wind, fuel, terrain, and suppression, whereas a coordinated public-safety incident evolves through human decisions, mobility, communication, and response. Third, incidents may be *compositional*: multiple local observations or activities may need to be interpreted jointly as parts of the same incident, while unrelated anomalies must not be incorrectly merged.

We address this problem with *IncidentLens*, a sensing system for turning streams of local observations into event-level incident records. *IncidentLens* represents each candidate explanation as an incident hypothesis containing a possible incident type, origin region, propagation behavior, and set of supporting or contradicting observations. Given heterogeneous observations, *IncidentLens* generates and maintains competing hypotheses that could explain the

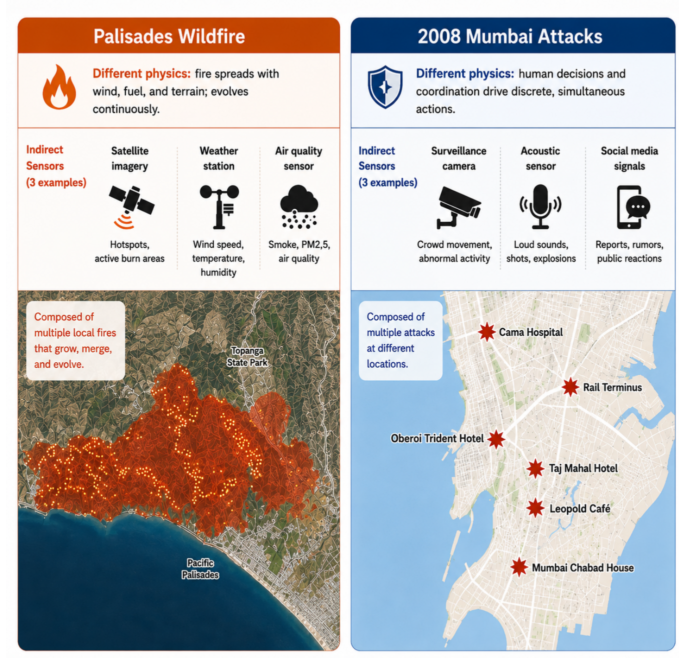

Figure 1: Motivational example of different incident characteristics

observed effects, scores them according to their agreement with incoming evidence, and updates or decays hypotheses as new observations arrive. This structures city-scale incident detection as latent-cause inference under sparse, indirect, and heterogeneous observability: the system reasons from distributed effects back to candidate incident sources and produces records that explain why local observations belong together.

This paper makes the following contributions:

- We formulate city-scale incident detection as latent source-hypothesis maintenance, where distributed observations are interpreted as indirect effects of hidden incident sources rather than as independent local anomalies.
- We design *IncidentLens*, a streaming system that maintains competing incident hypotheses over space and time, using incident-specific source priors, propagation models, and clustering criteria within a shared online inference engine.
- We evaluate *IncidentLens* on real Los Angeles data and controlled synthetic incident scenarios, comparing against direct observation, space-time clustering, text-only, late-fusion, and hotspot-scan baselines.
- We plan to release the implementation, evaluation code, synthetic generation pipeline, and shareable real-data labels/metadata to support reproducibility and future research in city-scale incident detection.

2 Problem Setting

2.1 Incident Definitions

Incidents are defined under the U.S. National Incident Management System (NIMS) as "an occurrence or event, natural or human-made, that necessitates a response to protect life, property or the environment" [17]. In this work, we focus on unplanned incidents, such as emergencies and disasters, rather than planned events (such as concerts or parades). NIMS/ICS further characterizes incidents using a five-level complexity scale, ranging from Type 5 incidents, which are the least complex, to Type 1 incidents, which are the most complex [19, 20]. Although these types are not defined by fixed spatial or temporal thresholds, higher-complexity Type 1–2 incidents typically extend across multiple operational periods, require substantial external resources, and may involve multijurisdictional or multidisciplinary response. In contrast, Type 3–5 incidents are generally more localized, shorter in duration, and require fewer response resources.

From the perspective of our system, we distinguish between three incident scales, summarized in Table 1. *Small-scale incidents* refer to lower-complexity incidents, roughly corresponding to NIMS/ICS Type 3–5 incidents, whose effects are expected to have a limited spatial and temporal footprint. Such incidents are therefore likely to be observable by only a few sensory vantage points. *Large-scale incidents* refer to higher-complexity Type 1–2 incidents, whose broader spatial and temporal footprints are more likely to generate distributed observations across many sensors, regions, and modalities. Finally, *incident complexes* refer to the NIMS-defined grouping of two or more individual incidents in the same general area managed by a single Incident Commander or Unified Command [18].

These categories describe sensing footprint and organizational treatment rather than mutually exclusive causal classes. A small-scale incident may be the origin or early manifestation of a large-scale incident, such as a small ignition that develops into a wildfire or a localized road closure that later becomes part of a broader evacuation response. Conversely, an incident complex may compose multiple individual incidents, including small-scale incidents, large-scale incidents, or a mixture of both, when they occur in the same general area and are managed under a shared command structure. Thus, our terminology captures the difference between isolated small-scale incidents, large-scale incidents with broad sensing footprints, and composite incidents whose evidence arises from multiple related sub-incidents. In this work, we focus on detecting unplanned emergencies and disasters whose sensing footprints correspond to large-scale incidents, incident complexes, or large-scale incidents that emerge from initially small-scale incidents.

2.2 Problem Formulation

A city is instrumented with data sources $\mathcal{D} = d_1, \ldots, d_M$, including traffic sensors, air-quality monitors, cameras, operational text reports, social media posts, and official alerts. Each raw input record is

$$r_i = (d_i, x_i, t_i, y_i),$$

where d_i is the data source, x_i is its location or spatial footprint, t_i is its timestamp, and y_i is the raw payload. The city is represented by grid cells $\mathcal{G} = g_1, \ldots, g_N$ with static and historical context $\mathcal{X}$, such as roads, land cover, terrain, sensor coverage, and historical baselines.

The goal is to infer incident records from streams of partial observations. We represent an incident as a latent spatiotemporal event

$$E_k = (c_k, S_k, A_k(t), I_k, \phi_k),$$

where c_k is the incident type, S_k is the source region, $A_k(t)$ is the affected region over time, $I_k = [t_k^s, t_k^e]$ is the active interval, and ϕ_k is its internal state. Incidents are latent because the source event is often not directly observed. Instead, the system observes effects that may be displaced in space or time, such as smoke downwind of a fire, congestion downstream of a crash, water near a flood, or public reports after an emergency response has begun.

This creates four spatial quantities that may not coincide: the *reported location* named by an external source, the *observed-effect location* where evidence is sensed, the *inferred source region* estimated by the system, and the *affected region* where effects are observed or predicted. For example, a wildfire may originate in a hillside cell while smoke, degraded air quality, evacuation reports, and congestion are observed kilometers away. This distinction is central to our problem setting: incident detection requires explaining distributed effects while estimating both the latent source and the region affected over time.

Given raw inputs up to time T, denoted $\mathcal{R}_{[0,T]}$, the system outputs a ranked set of incident predictions

$$\widehat{\mathcal{E}}_T = (\hat{c}_k, \hat{S}_k, \hat{A}_k(t), \hat{I}_k, \hat{\phi}_k, \hat{p} * k) * k = 1^K.$$

Here $\hat{p}_k$ is a relative support score rather than a calibrated posterior probability. The number of active incidents K is unknown in advance: new incidents may appear, multiple observations may refer to the same incident, and some observations may not correspond to any incident in the target scope. The task is therefore not only to classify individual observations, but to decide how many incident records should be created, where their latent sources may be, and which observed effects they explain.

Composition is treated as a secondary capability. Some records may later be linked as part of a broader incident, such as multiple fires associated with a regional wildfire or several local public-safety records associated with a larger event. The primary problem in this paper is to infer event-level incident records whose evidence can be explained by observable effects in the available data streams. *IncidentLens* is not intended to replace specialized wildfire simulators, traffic crash classifiers, gunshot localization systems, or other incident-specific detectors; it addresses the systems problem of maintaining incident-level predictions when evidence is sparse, indirect, multimodal, and weakly labeled.

2.3 Noisy Labels and Event-Level Evaluation

Real incident labels are delayed, incomplete, and biased toward reported events. A label may contain only a category, coarse location, approximate time interval, and provenance. We represent such an annotation as

$$\tilde{E}_k = (\tilde{c}_k, \tilde{A}_k, \tilde{I}_k, \tilde{m}_k),$$

where $\tilde{c}_k$ is the reported incident category, $\tilde{A}_k$ is the reported or affected area, $\tilde{I}_k$ is the approximate time interval, and $\tilde{m}_k$ stores metadata such as publication time and label provenance.

Because exact latent state recovery is usually unidentifiable in real data, we evaluate compatibility with available labels rather than closed-world correctness. A strict event-level match requires

Table 1: Sensing-oriented incident terminology grounded in NIMS/ICS incident complexity.

Term	NIMS/ICS grounding	Sensing interpretation	Examples
Small-scale incident	Paper-specific term for lower-complexity incidents, roughly corresponding to Type 3–5 incidents on the NIMS/ICS complexity scale [19, 20].	Limited spatial and temporal footprint; likely to be observed by one sensor or a small cluster of nearby sensors.	Home crime; small fire; local road closure.
Large-scale incident	Paper-specific term for higher-complexity incidents, roughly corresponding to Type 1–2 incidents, which involve multiple operational periods, substantial resources, and potentially multijurisdictional or multidisciplinary response [19, 20].	Broader spatial and temporal footprint; likely to produce distributed observations across many sensors, regions, and modalities.	Wildfire; earthquake; major flood.
Incident complex	NIMS-defined grouping of two or more individual incidents located in the same general area and managed by a single Incident Commander or Unified Command [18].	Composite incident whose evidence may appear as multiple related spatial or temporal clusters that should be interpreted jointly.	Multiple nearby wildfires; related explosions and fires; storm-related closures and rescues.

semantic, spatial, and temporal compatibility:

$$\text{match}_{\text{strict}}(\hat{E},\tilde{E}) = \mathbf{1}[\text{sim}_c(\hat{E},\tilde{E}) \geq \delta_c \wedge \text{sim}_x(\hat{E},\tilde{E}) \geq \delta_x \wedge \text{sim}_t(\hat{E},\tilde{E}) \geq \delta_t].$$

The similarity functions are instantiated according to label granularity: type hierarchy agreement or canonical label equality for sim_c, distance or spatial overlap for sim_x, and interval overlap for sim_t.

Synthetic labels provide source and affected-region ground truth, so synthetic evaluation can apply the full semantic, spatial, and temporal match. Real labels are weaker: they may describe reported or affected locations rather than true sources. For this reason, the real-data evaluation treats type/time matching as the primary weak-label event metric and reports spatial quality separately. These real-data metrics measure compatibility with available reported labels, not complete latent source recovery or a closed-world census of all incidents.

3 Methodology

3.1 Overview and Terminology

IncidentLens separates two responsibilities that are often entangled in multimodal incident detection. The first is *semantic extraction*: converting modality-specific raw inputs into a common evidence representation. The second is *spatiotemporal hypothesis maintenance*: maintaining a persistent online state of possible incident sources that could explain those observations over space and time.

Figure 2 shows the system architecture. Offline, *IncidentLens* constructs a city grid, indexes static context layers, and validates incident-function specifications for each incident type. Online, each raw input record is mapped to a semantic observation. The hypothesis engine updates existing source hypotheses, proposes new hypotheses by reasoning backward from observed effects to possible causes, prunes low-support hypotheses, and passes stable hypotheses to incident discovery. Incident memory is used after discovery to maintain temporal consistency, suppress duplicate reports, and conservatively link records when there is explicit evidence that they are part of a larger incident. Thus, the core detection problem is source-hypothesis maintenance; composition is not required for an incident record to be detected. Figure 3 illustrates this observation-to-incident flow.

A raw input record is a timestamped sensor, camera, text, or alert item with a location or footprint. The observation extractor maps it to a semantic observation $o_i = (x_i, t_i, m_i, E_i, I_i, q_i)$, where E_i contains observed effects and I_i contains candidate incident types. A source hypothesis is a candidate latent cause with incident type, source cell, start time, propagation parameters, and support score. Incident-function specifications provide the incident-type-specific source priors, propagation models, clustering distances, and bounded parameters used by the shared online engine. Stable clusters of source hypotheses are emitted as incident records. Each emitted record therefore separates the inferred source region from the affected region and from the locations where evidence was observed.

When evaluating on historical data, we process records in timestamp order to simulate online operation. We refer to this as *stream replay*: past records are fed to the online system as if they were arriving live.

3.2 Semantic Observation Extraction

The Observation Extractor maps each raw input record to

$$o_i = (x_i, t_i, m_i, E_i, I_i, q_i),$$

where x_i is a location or footprint, t_i is time, m_i is modality, E_i is a sparse set of ontology effects, I_i is a set of candidate incident types, and q_i is a bounded quality score. Each effect is (ℓ, ρ) with strength $\rho \in [0, 1]$. This representation is evidence, not a final incident decision. Each effect label ℓ is drawn from a fixed semantic effect ontology; representative effects, supported modalities, and extraction examples are listed in Appendix A.2.

Extraction has a lightweight anomaly stage followed by schema-bounded semantic parsing. The anomaly stage filters or prioritizes reports using modality-specific signals such as text matches, visual prefilters, and structured time-series anomalies. The observation model then emits only fixed-vocabulary effects, candidate incident

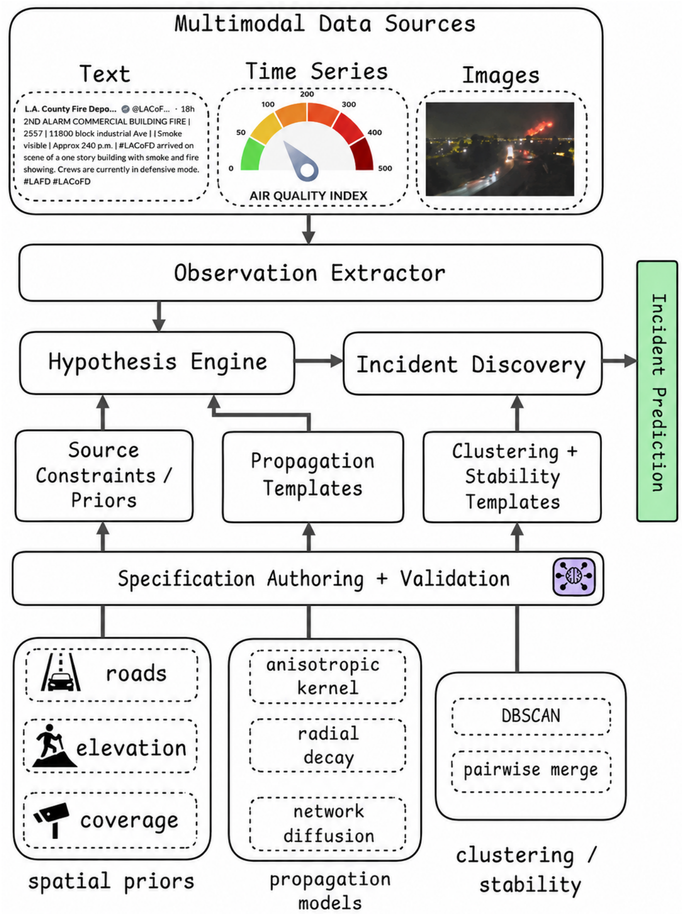

Figure 2: System architecture of *IncidentLens*. Semantic observations and incident-type specifications feed a shared online engine that maintains source hypotheses and emits incident records.

types, locations, timestamps, and quality scores. Context-only variables such as wind or temperature may condition later hypothesis updates but do not create incidents by themselves.

The extractor is hypothesis-aware only as a gating mechanism: reports inside the predicted support of an active hypothesis may bypass the anomaly prefilter so weak follow-up evidence is not discarded. Hypothesis context does not assign labels, update weights, or create incident records without an emitted semantic observation. Foundation models, when used, operate only behind the closed schema and are not allowed to emit free-form incident records or consume label-revealing metadata.

3.3 Incident-Function Specifications

The main interface between incident-specific reasoning and the shared online engine is the incident-function specification. Each incident type c is associated with

$$F_c = (Q_c, P_c, D_c, \Theta_c),$$

where Q_c is a bounded source-prior score, P_c is a set of effect-specific propagation models, D_c is a clustering distance, and Θ_c contains bounded parameters. This interface lets incident types express different physical and semantic assumptions while using the same online proposal, update, pruning, clustering, and stability logic.

The source prior $Q_c(g \mid \mathcal{X})$ is a proposal score over feasible source cells, not a probability that an incident is present at g. Hard constraints remove physically invalid cells, such as road incidents away from the road network, while soft context features are clipped and normalized within the current reverse-proposal envelope. Thus, Q_c affects which source hypotheses are proposed under a fixed budget, but it cannot by itself create an incident record. We also distinguish source plausibility from observability: sensor coverage and data availability determine whether a modality can support or contradict a hypothesis, but sensor density is not treated as evidence that an incident occurred.

For each effect ℓ, the propagation component provides a model

$$C_\ell(g, t \mid h_j, \mathcal{X}) \to [0, 1],$$

which estimates how strongly effect ℓ should appear at location g and time t under source hypothesis h_j. The clustering distance D_c defines when hypotheses of the same incident type are similar enough to form an incident candidate, using terms such as source distance, start-time difference, affected-region overlap, predicted-effect overlap, and shared supporting observations. The parameter set Θ_c bounds propagation scale, lookback horizon, effect weights, proposal budget, pruning thresholds, clustering thresholds, and stability thresholds.

Specifications are schema-bounded templates rather than arbitrary per-incident programs. They select source constraints, propagation families, clustering terms, and parameter ranges from a fixed catalog. Foundation models may assist with offline or lazy template authoring, but their outputs must be structured and validated: they cannot introduce new effect labels, arbitrary propagation code, report-specific coordinates, or incident records. Runtime values such as source cell and start time are bound only when a hypothesis is instantiated. Appendix A.3 gives example specifications for wildfire, road accident, flood, and crowd disruption.

3.4 Streaming Source-Hypothesis Update

IncidentLens maintains a bounded set of source hypotheses

$$h_j = (c_j, s_j, \tau_j, \theta_j, M_j, w_j),$$

where c_j is incident type, s_j is source cell, τ_j is start time, θ_j contains bounded runtime parameters, M_j is the bound effect-model set, and w_j is a relative support score. This representation is inspired by multi-hypothesis tracking and particle filtering, but the weights are not calibrated posterior probabilities: they are online support scores for competing incident explanations under sparse, displaced, and multimodal evidence.

The streaming update follows a birth–update–prune pattern similar in spirit to multi-hypothesis filtering, but specialized to incident explanations rather than physical targets. When a semantic observation o_i arrives, *IncidentLens* first scores existing hypotheses before proposing new ones. This ordering prevents an observation from immediately reinforcing hypotheses that it created. Existing hypotheses receive positive support when their predicted effects agree with the observation and negative support when a strongly observed effect is inconsistent with their predicted affectedness. New hypotheses are then proposed only for candidate incident types and source regions whose forward propagation models could plausibly explain the observation.

For each active hypothesis h_j and each observed effect $(\ell, \rho_{i,\ell}) \in E_i$, the engine computes the predicted affectedness

$$a_{j,\ell} = C_\ell(x_i, t_i \mid h_j, \mathcal{X}).$$

Hypothesis support is updated with

$$w'_j = w_j \exp(\eta q_i u_j(o_i)),$$

where η is an update scale, $q_i \in [0, 1]$ is a bounded observation-quality weight, and $u_j(o_i) \in [-1, 1]$ is a clipped compatibility score. The weights are relative online support scores, not calibrated posterior probabilities. They are used to rank, prune, diversify, and cluster hypotheses; incident records are emitted only after persistent cluster-level support.

The compatibility score rewards predicted effects that are observed, gives a small bonus when predicted and observed strengths are close, penalizes strong observed effects in regions where the hypothesis predicts little affectedness, and adds weak support when the observation explicitly names the same incident type. Effects without a corresponding propagation model do not contribute positive evidence. Missing modalities are not treated as negative evidence: an unreported camera, traffic, or text effect does not contradict a hypothesis unless an explicit observed effect is incompatible with the hypothesis prediction.

The quality weight q_i controls the influence of noisy observations. It combines source or modality reliability, localization precision, extraction specificity, and anomaly strength when available, with missing components treated as neutral. Thus, q_i can down-weight weak or ambiguous reports, but it cannot create a hypothesis or incident record by itself.

New hypotheses are proposed by reverse-envelope search. For each candidate incident type $c \in I_i$, observed effect ℓ, and discrete lookback start time τ, the engine enumerates source cells g whose forward model could plausibly explain the observation:

$$C_\ell(x_i, t_i \mid c, g, \tau, \mathcal{X}) \geq \epsilon_{\text{rev}}.$$

Cells outside this envelope are ignored. The remaining cells are filtered by hard source constraints, reweighted by the bounded source prior Q_c, and sampled under a fixed proposal budget with spatial diversity. Algorithm 1 summarizes the per-observation streaming update. Each sampled cell and start time is then bound to the incident-type propagation template to create a source-hypothesis particle.

Some streams are evidence-only rather than proposal sources. For example, weather can strengthen or weaken existing wildfire hypotheses, but it should not by itself create a wildfire source hypothesis. Such support-only sources are used during hypothesis evaluation but skipped during hypothesis birth. This prevents contextual variables from generating incident detections without a concrete observed effect.

Algorithm 1 Streaming source-hypothesis update

Require: source hypotheses H_t, semantic observation o_i, specifications $\mathcal{F}$, cap $N_{\max}$

for all $h_j \in H_t$ **do**
  Score predicted effects with $C_\ell(x_i, t_i \mid h_j, \mathcal{X})$
  Update $h_j.w \leftarrow h_j.w \exp(\eta q_i u_j(o_i))$
end for
for all $c \in I_i$ and $(\ell, \rho) \in E_i$ **do**
  Load $F_c = (Q_c, P_c, D_c, \Theta_c)$
  Compute bounded reverse-support envelope for (ℓ, ρ)
  Enumerate feasible source cells and start times
  Apply source constraints, reweight cells with Q_c, and sample diverse particles
end for
Normalize weights
Merge or suppress duplicate hypotheses
Prune low-support hypotheses and enforce diversity under $N_{\max}$
return updated hypothesis set H_t^+

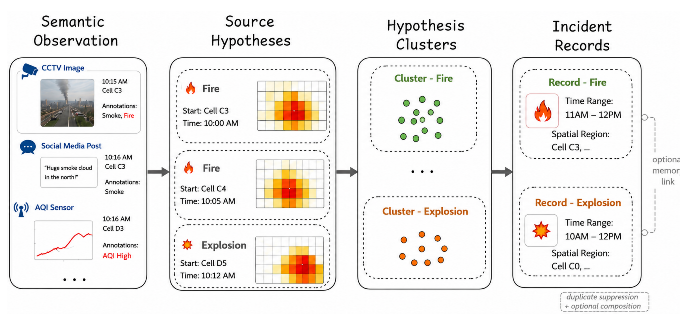

Figure 3: Observation-to-record pipeline. Semantic observations update source hypotheses, stable hypothesis clusters become incident records, and incident memory handles duplicate suppression and optional composition.

After scoring and proposal, weights are normalized. Duplicate hypotheses are merged or suppressed when they have the same incident type, nearby source cells, similar start times, and overlapping effect models. Low-support hypotheses are pruned, and a stratified top-k selection preserves diversity across incident type, source region, and start time. This maintained hypothesis population is the core online state of *IncidentLens*.

Soft data association. *IncidentLens* uses soft many-to-many association: each observation updates all active hypotheses through the compatibility score rather than being assigned to a single track. Event-level association is deferred to clustering, where hypotheses of the same incident type are grouped by source distance, start time, effect overlap, and supporting-report overlap.

3.5 Incident Discovery and Stability

Periodically, *IncidentLens* clusters source hypotheses of the same incident type using D_c. Each cluster defines a candidate incident

$$R_k = (\hat{c}_k, \hat{S}_k, \hat{\tau}_k, \hat{A}_k(t), M_k, U_k),$$

where $\hat{c}_k$ is type, $\hat{S}_k$ is source region, $\hat{\tau}_k$ is start time, $\hat{A}_k(t)$ is aggregate affectedness, M_k is normalized cluster mass, and U_k summarizes supporting reports, sensors, effects, and locations. Candidates are matched across update rounds by type, source region, and start time so that small particle changes do not produce duplicate records.

A candidate is promoted only after satisfying stability thresholds on cluster mass, age, recent support, decay, and missed updates. Stable records are emitted as

$$\hat{E}_k = (\hat{c}_k, \hat{S}_k, \hat{A}_k(t), \hat{I}_k, \hat{\phi}_k, \hat{p}_k, O_k),$$

where $\hat{I}_k$ is the inferred active interval, $\hat{\phi}_k$ is state, $\hat{p}_k$ is a support score, and O_k is the supporting evidence summary. Here $\hat{S}_k$ denotes the inferred source region, while $\hat{A}_k(t)$ denotes the affected region. These are distinct from the observed-effect locations in the supporting evidence summary O_k and from any reported location in an external label.

LLM leakage mitigation. When an LLM is used for bounded cluster assessment, it receives only de-identified structured summaries: relative local coordinates/times, support counts, modality/effect categories, stability features, and neighborhood consistency. Prompts omit exact dates, coordinates, place names, incident names, URLs, report IDs, sensor IDs, and agency names, and labels are restricted to the allowed ontology. This cannot remove all pretrained world knowledge, but it reduces cues that would allow the model to retrieve memorized real-world events rather than reason from replayed evidence.

3.6 Incident Memory and Composition

Incident memory is a post-discovery module used to maintain temporal consistency and suppress duplicates. When a stable record is emitted, it is compared with recent records of compatible type, source or affected region, active interval, and supporting evidence. Compatible records update the existing memory entry rather than producing duplicate reports.

Composition is optional and conservative. After event-level records already exist, memory may create a *part-of* link when heterogeneous incidents have explicit evidence of a broader event, such as fire plus road closure or shooting plus bomb threat. These links do not create the initial low-level detections and do not increase their detection scores in the main evaluation. More details found in Appendix A.

Use of AI tools. We used generative AI tools during parts of the research workflow. AI assistance was used to help draft and refactor portions of the system codebase, implement evaluation scripts and baseline methods, and debug experimental infrastructure. Generative AI was also used by design in the synthetic-data pipeline, including image modification and generation of incident-relevant synthetic observations such as text and time-series effects. These generated artifacts are part of the experimental setup and are described as synthetic data throughout the paper.

4 Evaluation

The evaluation tests one central claim: maintaining source-level hypotheses improves event-level detection when observations are sparse, indirect, and multimodal. We treat localization, timeliness, composition, and runtime as supporting properties rather than equally strong claims, because their observability differs between synthetic and real data.

We first describe the datasets, labels, metrics, and baselines. We then report synthetic low-level detection, real-world weak-label detection, focused ablations, synthetic composition, and stress/runtime results. The synthetic experiments provide the cleanest test of source and affected-region recovery because ground truth is known. The real-data experiments test whether the same design improves detection against weak external incident records, with spatial quality reported as a diagnostic rather than as the primary matching criterion. Implementation details are described in Appendix B.1

4.1 Datasets and Labels

Label isolation and text provenance. For both the synthetic and real-data experiments, label-construction data are used only for evaluation and are never provided as input to *IncidentLens* or to any baseline. In the real dataset, news articles are used to construct incident labels, but the replay stream contains only non-news observations, including sensors and operational text sources such as social posts, citizen reports, and official alerts. In the synthetic dataset, incident labels are derived from simulator plans and ground-truth metadata, but these plans and labels are not exposed to any detector during replay. Thus, in both settings, the systems are evaluated against held-out labels whose provenance is disjoint from the observation streams used for detection. The synthetic ablations further separate physical-sensor and operational-text performance.

Real-world data. We evaluate on Los Angeles observations from January 2025 to May 2026. Replay inputs include Caltrans CCTV [2], ALERTCalifornia cameras [1], PurpleAir air quality[6], OpenWeatherMap weather [4], Caltrans PeMS traffic [10], CitizenApp alerts [3], and verified public-safety accounts on X [59]. GDELT/news metadata [5] is used only for label construction and is not replayed to *IncidentLens* or to any baseline. The dataset has intermittent missing intervals and unequal temporal coverage across sources, reflecting realistic heterogeneous sensing conditions. Appendix C gives the full source and dataset statistics.

Weak-label construction. Labels are constructed offline from external reports using an LLM-assisted parser and deduplicated by semantic, spatial, and temporal overlap. We treat them as weak records of reported incidents rather than complete ground truth; Appendix D describes the labeling pipeline.

Synthetic data. Synthetic incidents provide controlled multimodal scenarios with complete source-region, affected-region, timing, and provenance ground truth. The synthetic dataset contains 197 singular cases and 97 composite cases, spanning 294 simulator runs and 391 localized incidents. Appendix C gives the full source and dataset statistics.

Each synthetic case is grounded in a Los Angeles-area region and emits incident-relevant observations from edited CCTV / ALERTCalifornia imagery, structured time series such as air quality, weather, and traffic speed/occupancy, and textual public-report-like sources such as news, citizen reports, and social media. We use the synthetic dataset for settings where real-world ground truth is difficult to obtain at the required granularity, including stress tests over observability, sensor dropout, noise, indirect-effect displacement, and missing modalities. Although the simulator and detector share a broad incident/effect ontology, the detector receives only emitted observations and sensor metadata, not simulator state, hidden incident regions, generation prompts, or ground-truth labels. For more information on how we generated this synthetic dataset, see Appendix B.

The synthetic benchmark is intended to test controlled observability rather than to fully replace real deployment evidence. Because the simulator and detector necessarily share a broad incident/effect ontology, we interpret synthetic results as evidence that the hypothesis-maintenance design works under known latent incidents, not as proof that the exact templates will transfer unchanged to all cities or incident families. We reduce direct circularity by withholding simulator plans, hidden incident state, ground-truth regions, generation prompts, and labels from all detectors during replay.

4.2 Metrics and Matching Protocol

For event-level detection, we report precision, recall, F_1, F_2, and false positives. We emphasize F_2 because missed incidents are more costly than a small number of extra candidates, while false positives are still penalized. Synthetic evaluation uses one-to-one matching between predictions and simulator labels. A correct synthetic match requires compatible incident type, temporal overlap, and spatial compatibility, following the noisy-label matching protocol from Section 2.3. A wrong-type match is counted as a false positive for the predicted type and a false negative for the ground-truth type. We distinguish four spatial quantities throughout the evaluation: *reported location* is the location named by an external label or report; *observed-effect location* is where evidence is sensed; *inferred source region* is the latent origin estimated by *IncidentLens*; and *affected region* is the region where effects are observed or predicted. Synthetic labels contain source and affected-region ground truth, while real labels often contain reported or affected locations rather than true sources.

Spatial metrics are reported in two forms. First, localization error is the centroid distance between the predicted inferred source region and the ground-truth or reported label region. For methods that do not estimate an explicit source, we use the predicted affected region or support points as a source-location proxy. Second, scenario-level spatial metrics compare the union of all same-type predicted affected regions in a run against the corresponding ground-truth affected region. We report spatial coverage recall,

$$\mathrm{SRec} = \frac{|\hat{A} \cap A|}{|A|},$$

and spatial intersection-over-union,

$$\mathrm{SIoU} = \frac{|\hat{A} \cap A|}{|\hat{A} \cup A|}.$$

Thus, S rec. measures whether the method covers the true affected region, while S IoU also penalizes over-predicted area. Temporal IoU is computed analogously by merging same-type predicted and ground-truth active intervals before computing intersection-over-union. For each ground-truth incident type in a run, we union all same-type predicted regions and compare that union against the corresponding ground-truth region.

For real data, labels are weak external incident records rather than complete simulator ground truth. The primary real-data results therefore use the configured real matching policy: predictions are matched to labels by canonical incident type and temporal compatibility, with optional spatial gating when cached label geometries are available. We pool predictions across date folders before matching because real incidents can span multiple days. Timeliness is measured relative to the earliest external article timestamp,

$$\mathrm{Delay}(\hat{E}, \tilde{E}) = t_{\mathrm{detect}}(\hat{E}) - t_{\mathrm{article}}(\tilde{E}),$$

where negative values indicate detection before the article used for label construction. We also report pre-article recall (Pre-rec.), the fraction of all labels matched by a detection before the earliest article timestamp, and pre-article fraction (Pre-frac.), the fraction of matched true positives detected before that timestamp.

Because source availability places an upper bound on what can be observed, we also compute an incident-level *observability coverage* score. Coverage is computed from normalized source inventories rather than from prediction support: the synthetic run uses simulator-emitted source reports, and the real run uses the real-emitter source inventory. For an incident e, the score combines modality suitability, source availability near the incident in space/time, and weak inferential evidence availability:

$$C(e) = w_m C_{\mathrm{mod}}(e) + w_s C_{\mathrm{src}}(e) + w_i C_{\mathrm{inf}}(e).$$

We report coverage-weighted recall,

$$R_C = \frac{\sum_{e \in GT} C(e) \mathbf{1}[e \text{ matched}]}{\sum_{e \in GT} C(e)},$$

as a diagnostic showing whether a method mainly succeeds on highly observable incidents. Appendices F and G give the coverage distributions, coverage-bin results, and correlation analyses; Appendix F.1 reports a controlled smoke test for the coverage computation itself.

Composition metrics are reported separately. In synthetic data, the composition table evaluates positive composition scenarios, so we interpret its detection metrics as positive-case composition recall and its type metric as the accuracy of the predicted high-level incident type among detected true positives. In real data, the available top-level labels are broad incident families rather than complete high-level relation graphs, so we report weak top-level support: whether a method detects enough listed low-level child incidents to support a top-level family.

4.3 Baselines

Baselines. We compare against five streaming baselines that represent common alternatives to source-hypothesis reasoning. *Direct observation* emits an incident whenever a single semantic observation exceeds a confidence threshold, testing whether incident detection can be solved by local report classification alone. *Space-time clustering* links same-type semantic observations within a fixed spatial and temporal radius, providing a non-parametric clustering baseline for repeated local evidence. *Text-only clustering* applies the same idea only to operational text streams, isolating the strength of text without sensor fusion. *Hotspot scan* is our strongest general

Method	Prec. ↑	Rec. ↑	F_1 ↑	F_2 ↑	FP/run ↓	S rec. ↑	S IoU ↑	T IoU ↑
IncidentLens	**0.438**	**0.762**	**0.556**	**0.664**	**0.98**	**0.723**	0.064	**0.541**
SaTScan bg.	0.116	0.370	0.176	0.257	2.83	0.359	0.068	0.248
Hawkes detector	0.142	0.381	0.207	0.285	2.30	0.369	0.069	0.213
Direct obs.	0.006	0.630	0.013	0.031	96.76	0.374	**0.153**	0.303
ST clustering	0.043	0.619	0.081	0.169	13.73	0.503	**0.153**	0.301
Late fusion	0.038	0.564	0.071	0.150	14.25	0.515	0.146	0.277
Hotspot scan	0.030	0.569	0.057	0.123	18.56	0.564	0.080	0.281
Text only	0.041	0.519	0.076	0.155	12.20	0.288	0.087	0.227
Generic propagation	0.000	0.000	0.000	0.000	1.31	0.000	0.000	0.000

Table 2: Aggregate synthetic low-level incident detection results. Event metrics are micro-averaged; S rec., S IoU, and T IoU use scenario-level union accounting.

Condition	Sources	Mean C	Pred.	TP	FP	Recall	F_2
None	0	0.000	0	0	0	0.00	0.000
Low	90	0.591	5	5	0	0.50	0.556
Medium	314	0.609	11	10	1	1.00	0.980
High	807	0.609	14	10	4	1.00	0.926

Table 3: Synthetic coverage smoke test over ten replayed incidents. Coverage is computed from the emitted source inventory, not from prediction support.

classical baseline: it searches for local concentrations of same-type semantic evidence over grid/time windows, capturing the main intuition behind scan-statistic and hotspot detectors while remaining applicable to sparse heterogeneous reports. Finally, *late-fusion voting* is included as a conservative multimodal-agreement diagnostic: it emits an incident only when multiple modalities support the same incident type in a local space-time window. Its role is to test whether simple cross-modal agreement is sufficient, not to represent the strongest possible detector.

We do not include full inverse-model baselines or a literal Kulldorff SaTScan implementation as primary comparators because they are not uniformly applicable to our setting. Inverse methods such as wind-based smoke/PM2.5 back-projection or traffic-network back-tracing require incident-specific physics, calibrated auxiliary variables, and modality-specific assumptions; they are valuable for particular incident families, but do not naturally cover the full ontology of fires, floods, road disruptions, protests, public-safety threats, and composite incidents. Similarly, Kulldorff-style SaTScan assumes a formal spatial or space-time scan statistic over count or case-control streams with an explicit background model, whereas our observations are sparse semantic outputs from heterogeneous sensors and operational text. We therefore use the hotspot scan as the closest general-purpose scan-statistic-style comparator and treat inverse-model baselines as important incident-specific extensions.

All baselines use the same set of input data and the same event-level matching protocol. When a baseline consumes semantic observations, it receives the same extractor outputs as *IncidentLens*. Thresholds are selected on the same validation period. Text-only baselines receive only text available by replay time t, and generic-propagation baselines use the same particle budget as *IncidentLens*.

4.4 Synthetic Low-Level Detection Results

Figure 4 breaks down micro-F_2 for selected synthetic incident categories. The figure shows that *IncidentLens* is strongest on flood, wildfire, protest, and urban-fire scenarios, while terrorist-attack scenarios remain difficult for all methods.

Table 2 reports aggregate synthetic low-level incident detection. *IncidentLens* achieves the best event-level precision, recall, F_1, and F_2, while producing far fewer false positives per run than the direct, clustering, fusion, hotspot, and text-only baselines.

We also compute synthetic observability coverage from a simulator-emitted source inventory rather than from prediction support. The coverage inventory contains 42,867 normalized source reports, covers 181 low-level ground-truth incidents, and uses no type-default fallbacks. Coverage scores range from 0.000 to 0.757, with 44 low-, 94 medium-, and 43 high-coverage incidents. Coverage-weighted recall for *IncidentLens* is 0.808 versus 0.762 ordinary recall, and the Spearman correlation between coverage and whether an incident is matched is only 0.071. By contrast, direct observation and space-time clustering have much stronger coverage dependence (Spearman $\rho = 0.821$ and 0.801, respectively). This indicates that *IncidentLens* improves aggregate detection without simply shifting performance toward the most observable synthetic incidents. Appendix F reports the full synthetic coverage-bin and correlation analysis.

As a sanity check on the coverage computation, we additionally replay the same small set of ten supported synthetic incidents under four source-availability conditions: no emitted reports, low availability, medium availability, and full availability. Coverage is computed only from the emitted normalized REPORT inventory for each condition, with type-default fallbacks disabled. Table 3 shows the expected behavior: the no-coverage condition has zero source records, zero coverage, zero predictions, and zero recall/F_2. Increasing availability from low to medium raises both recall and F_2; the medium and high settings have the same mean coverage because the noisy-or coverage components saturate once enough relevant flood evidence is present.

4.5 Ablations

The ablations show that the most important contributors are reverse proposal, multimodal evidence, and incident-specific grouping. Removing reverse proposal causes the largest spatial degradation: S rec. drops from 0.798 to 0.172, indicating that proposing sources only near observed effects is insufficient when observations are indirect or displaced. The modality ablations also substantially degrade event-level performance, with F_2 dropping from 0.685 to 0.526 for sensor-only input and to 0.578 for operational text-only input, showing that neither modality family is sufficient across incident types. Generic clustering/stability also reduces F_2, suggesting that incident-specific grouping criteria help convert noisy hypothesis sets into stable incident records.

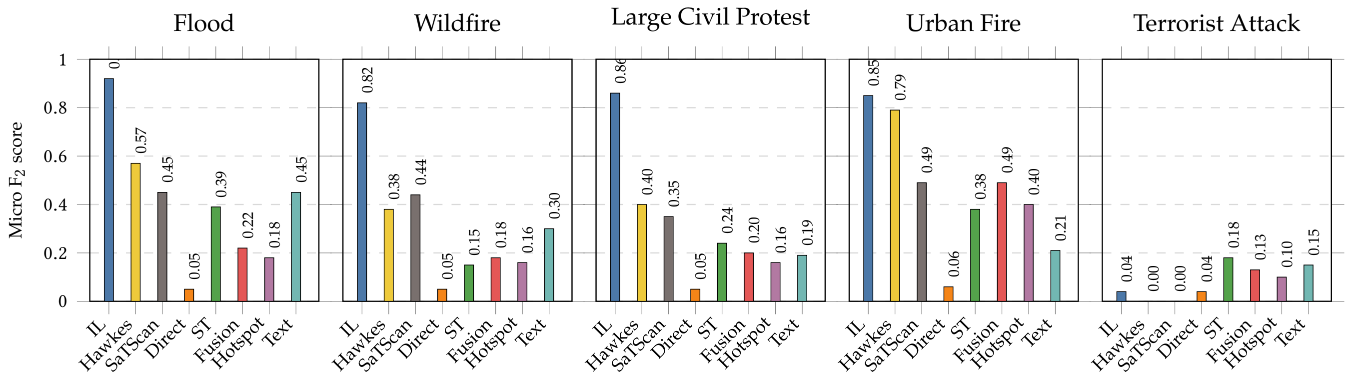

Figure 4: Scenario-conditioned micro-F$_2$ by selected ground-truth incident category on the synthetic low-level evaluation. Scores penalize extra predictions emitted in runs of the same ground-truth incident type.

Variant	F_2	ΔF_2	S rec.	ΔS rec.	T IoU	ΔT IoU
Full	0.685	–	0.798	–	**0.718**	–
Generic all	0.668	-0.017	0.806	+0.008	0.642	-0.076
Gen. cluster	0.644	-0.042	0.784	-0.014	0.714	-0.004
Gen. prop.	**0.690**	+0.004	**0.815**	+0.017	0.673	-0.045
No reverse	0.622	-0.064	0.172	-0.626	0.574	-0.144
Sensor only	0.526	-0.159	0.655	-0.143	0.592	-0.126
Text only	0.578	-0.107	0.562	-0.236	0.603	-0.115

Table 4: Synthetic low-level ablations. Delta columns report change relative to the full model.

Method	Detected	Rec.	F_2	Strict type
IncidentLens	**64/66**	**0.970**	**0.976**	0.938
Proximity linker	54/66	0.818	0.849	**1.000**
Type+time linker	47/66	0.712	0.756	0.000

Table 5: Synthetic incident-composition results over positive composition scenarios. Type accuracy is computed over detected true positives.

The propagation ablation is less conclusive in this dataset: generic propagation has similar F_2 and spatial recall to the full model, and in this run slightly improves both, although it lowers temporal IoU. We therefore do not interpret this ablation as independently validating incident-specific propagation templates. Instead, the strongest ablation evidence supports reverse source proposal, multimodal evidence, and incident-specific clustering/stability; propagation templates remain part of the system abstraction and appear most useful for temporal coherence.

4.6 Incident Composition

Incident composition. We also evaluate synthetic incident composition, where two lower-level sub-incidents may imply a higher-level incident such as wildfire, urban fire, civil protest, or terrorist incident. This task is not simply same-type linking: a composite event may combine heterogeneous children, such as fire plus road closure or shooting plus bomb threat. Table 5 compares *IncidentLens* with proximity-only and same-type temporal-overlap linkers on positive composition scenarios. The *proximity-only linker* predicts a composite incident whenever two lower-level incident records are within a fixed spatial distance, ignoring type and temporal overlap. The *same-type temporal-overlap linker* predicts a composite incident only when two lower-level records have the same incident type and overlapping active intervals. These results measure positive-case composition recall and high-level type accuracy, not the false-link rate on unrelated low-level incidents.

4.7 Stress Robustness and Runtime

Figure 5 reports stress robustness and controlled throughput. The first three panels show detection quality under missing observations, corrupted observations, and sensor-density stress settings on the 35-run scalability/stress subset. The throughput panel essentially duplicates reports for `wildfire1` with anomaly and observation-model processing enabled. M25/M50 remove 25%/50% of observations, C25/C50 corrupt 25%/50% of observations by replacing with a different effect/incident, and S50/S25 retain 50%/25% of sensors.

The throughput experiment is a controlled stress test rather than a full deployment benchmark: it duplicates one representative wildfire stream with anomaly and observation-model processing enabled. Table 6 shows that the symbolic hypothesis engine is not the dominant bottleneck; most time is spent in observation/anomaly processing and uninstrumented overhead. Deployment-scale throughput would therefore depend on filtering, caching, batching, parallelism, and the number of heavy model calls.

4.8 Real-World Detection Results

Table 7 reports aggregate real-data detection under the configured real-data matching policy. Under this weak-label protocol, *IncidentLens* achieves the best recall, F_1, F_2, pre-article recall, and coverage-weighted recall, while producing far fewer false positives than direct observation and the density-based baselines. Direct observation recovers fewer labels and emits more than five times as many false positives. The spatial and temporal clustering baselines, including SaTScan and Hawkes-style event detectors, detect only a small number of reported

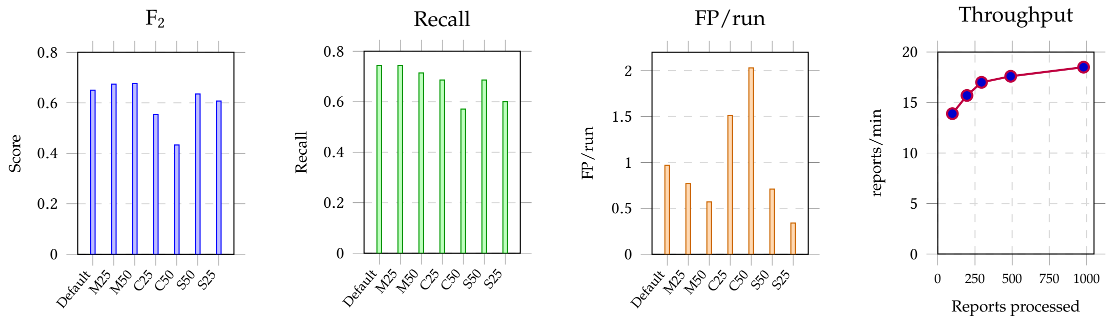

Figure 5: Stress robustness and throughput. The first three panels vary missing observations, corrupted observations, and sensor density; the final panel reports controlled end-to-end throughput.

Component	Time/report	Share
Hyp. proposal	0.029 s	0.8%
Hyp. evaluation	0.015 s	0.4%
Clustering	0.018 s	0.5%
Prediction	0.227 s	6.3%
Composition	0.055 s	1.5%
State/output write	0.075 s	2.1%
Obs./anomaly + overhead	**3.178 s**	**88.3%**
Total wall clock	3.598 s	100.0%

Table 6: Average per-report runtime breakdown for the controlled timing experiment with anomaly and observation-model processing enabled.

Method	Pred. ↓	Prec. ↑	Rec. ↑	F_1 ↑	F_2 ↑	FP ↓	TP	Pre-rec. ↑	C-rec. ↑
IncidentLens	96	**0.115**	**0.379**	**0.176**	**0.259**	85	**11**	**0.345**	**0.376**
Direct obs.	491	0.014	0.241	0.027	0.058	484	7	0.207	0.257
ST clustering	332	0.006	0.069	0.011	0.022	330	2	0.000	0.079
Hotspot scan	379	0.005	0.069	0.010	0.020	377	2	0.000	0.079
SaTScan bg.	153	0.007	0.034	0.011	0.019	152	1	0.000	0.042
Hawkes detector	44	0.023	0.034	0.027	0.031	43	1	0.000	0.042
Late fusion	6	0.000	0.000	0.000	0.000	6	0	0.000	0.000
Text only	0	–	0.000	0.000	0.000	0	0	0.000	0.000

Table 7: Aggregate real-data low-level detection over 29 weak labels under the configured real-data matching policy. C-rec. is coverage-weighted recall computed from a non-circular real-emitter source inventory.

incidents because the real observation streams are sparse, asynchronous, and often indirect. Late fusion obtains zero true positives because requiring local multimodal agreement is too strict under these conditions.

The coverage analysis uses 101,668 real-emitter source records and disables prediction-support fallback, so the coverage score estimates observability from the deployed source inventory rather than from any method's outputs. Coverage scores range from 0.518 to 0.749, with 23 medium-coverage and 6 high-coverage incidents. Coverage weighting leaves *IncidentLens*'s recall essentially unchanged (0.379 ordinary recall versus 0.376 coverage-weighted recall), whereas direct observation improves from 0.241 to 0.257 under coverage weighting. This suggests that direct observation is more concentrated on highly observable incidents, while *IncidentLens*'s gains are not explained by only detecting the easiest cases. Appendix G reports the full coverage-bin and correlation analysis.

We also evaluate weak real top-level incident support using the two broad top-level families available in the real labels: the Los Angeles wildfires and the Los Angeles anti-ICE protests. This is not a full graph-level composition benchmark; instead, it asks whether a method recovers enough listed low-level child incidents to support each top-level family. *IncidentLens* detects 8 of the 25 evaluable child incidents and supports one of the two top-level families, while direct observation detects 5 of 25 children and supports both families but does so with the much larger false-positive count shown in Table 7. The Hawkes, SaTScan, hotspot, and space-time clustering baselines each recover only one top-level child incident and support no top-level family. Appendix H reports the full child-support and linker-baseline diagnostics.

Appendix E reports an incident-level detection matrix for the named real labels, showing which methods detect each label and whether the detection occurs before the earliest external article timestamp.

Overall takeaways. Across synthetic low-level incidents, *IncidentLens* improves event-level precision, recall, and F_2 while reducing false positives by more than an order of magnitude relative to direct observation and simple clustering baselines. The spatial metrics are more nuanced: *IncidentLens* achieves high spatial recall but modest spatial IoU, indicating that it often covers the true affected region while also over-covering extra area.

The synthetic category breakdown reveals an important failure mode. Terrorist attack scenarios are difficult because the available observations often describe visible effects rather than intent or cause. Camera and sensor evidence may show fire, debris, smoke, road blockage, or emergency vehicles, while operational text may only describe response effects. In such cases, the evidence is often more directly compatible with urban fire, road disruption, or generic public-safety activity than with the causal label "terrorist attack."

On real data, *IncidentLens* detects the largest number of weak labels while reducing false positives by more than a factor of five relative to direct observation. Its coverage-weighted recall is nearly identical to its ordinary recall, suggesting that the gain is not simply due to finding only highly observable incidents. The weak top-level support results further show that real composition cannot yet be evaluated as a strict high-level graph task, but *IncidentLens* recovers more constituent child incidents than the clustering and scan baselines. The negative median report delay indicates that many matched detections occur before the earliest external article timestamp, supporting the provenance claim that news articles are used for labels rather than replayed as detector inputs.

Method	Loc. err. km ↓	S IoU ↑	T IoU ↑	Pre-frac. ↑	Med. delay h ↓	P25/P75 delay h
IncidentLens	**13.3**	**0.099**	0.149	**0.824**	**-29.2**	-48.7 / -4.2
Direct obs.	24.5	0.004	**0.205**	0.667	-17.3	-36.0 / 6.0
ST clustering	33.3	0.014	0.048	0.333	90.8	-1.9 / 367.4
Late fusion	–	0.037	0.000	–	–	– / –
Hotspot scan	30.4	0.005	0.045	0.200	38.3	8.8 / 165.8

Table 8: Real-data localization, overlap, and report-delay diagnostics under type/time non-spatial matching. Localization and delay are computed over matched true positives.

5 Discussion and Limitations

Discussion and limitations. *IncidentLens* is intended for observable incidents, not all urban events. Real-world labels are weak external records, and some may be outside sensing coverage, administrative rather than observable, or too underspecified to evaluate from available streams. Likewise, unmatched predictions may be false positives, duplicates, or plausible unreported incidents. We therefore interpret real-data precision and recall as measurements against reported labels, not as a complete census of city events.

We reduce label circularity by using news articles only for label construction; they are not replayed to *IncidentLens* or any baseline. Detector inputs are restricted to sensors and operational text such as social posts, citizen reports, and official alerts. This separation prevents direct reuse of label-construction articles, but does not remove all text-related bias: text-rich incidents may remain easier to detect than sensor-only incidents.

Foundation models introduce leakage and reproducibility risks. We mitigate these risks by using them only behind closed schemas for semantic extraction or bounded cluster assessment, and by excluding exact incident names, label articles, simulator ground truth, and other direct label cues from prompts. This reduces, but cannot eliminate, pretrained world-knowledge effects. Similarly, hypothesis weights are relative support scores for ranking, pruning, and clustering, not calibrated incident probabilities.

Synthetic data provides true source regions, affected regions, timing, and controlled observability, but may not capture the full ambiguity and reporting bias of real incidents. Some scenarios also expose limits of effect-based inference: terrorist-attack cases are difficult when observations show visible effects such as fire, smoke, debris, road blockage, or emergency response rather than intent or cause.

Finally, *IncidentLens* contains bounded design choices, including effect weights, propagation scales, source-prior weights, proposal budgets, pruning thresholds, clustering thresholds, and stability windows. Priors are locally normalized, proposals preserve spatial diversity, missing modalities are not treated as contradictions, and incident records require persistent cluster-level support. Deeper stress tests under deliberately misspecified priors, alternative propagation templates, and broader deployments remain future work.

Ethics and privacy. The system is intended to produce incident-level situational-awareness records, not person-level surveillance. The evaluation uses public or aggregated streams where possible, stores semantic effects rather than raw person identities, avoids tracking individuals across cameras or social media accounts, and reports only incident type, time, location, affected area, and supporting evidence. These choices reduce but do not eliminate surveillance risks; any deployment should include access controls, retention limits, audit logs, human oversight, and clear limits to incident-response or public-interest use.

6 Related Works

Urban sensing and anomaly detection. Urban sensing systems have long used fixed and mobile sensors to detect local anomalies in cities. Traffic systems detect freeway incidents from loop-detector occupancy and flow patterns using rule-based, neural, or probabilistic models [44, 46]; video systems detect accidents and anomalies in surveillance or dashcam streams [35, 43, 50, 53, 63]; and mobile sensing systems detect road hazards or crashes from vehicle- and phone-mounted sensors [16, 40, 58]. Other urban sensing work detects acoustic or social events, such as construction noise, sirens, gunshots, or local tweet bursts [11, 49, 67]. These systems are effective at recognizing that a particular sensor, road segment, camera view, trajectory, or modality is abnormal. In contrast, *IncidentLens* treats incidents as latent explanatory objects: observations may be displaced, delayed, modality-specific, or partial, and the core task is to infer which observations share a cause and where that cause likely originated.

Source localization and inverse urban sensing. Our work is also related to source localization, inverse modeling, and social sensing, which explicitly infer hidden causes from indirect observations. Prior work localizes pollutant releases using Bayesian inversion and dispersion models [13, 28], estimates wildfire smoke emissions and spread from satellite, weather, and atmospheric models [26, 47], detects disease or flood clusters from spatial reports [8, 30], localizes earthquakes or local events from social media [48, 66], and identifies shooters or traffic root causes from acoustic or mobility observations [12, 39, 51]. Recent learning-based systems further predict traffic accidents or risks over fixed spatial graphs using heterogeneous urban features [64, 65, 69]. These approaches reason backward from effects to causes, but typically assume the incident class, propagation mechanism, sensing graph, and relevant observations are known in advance. *IncidentLens* instead compares multiple incident hypotheses under uncertain event type, modality availability, spatial support, and observability.

Probabilistic association and spatiotemporal evidence fusion. IncidentLens is related to probabilistic tracking and data-association methods, which maintain uncertainty over latent states and ambiguous observations. MHT maintains competing measurement to track association histories [45], JPDA marginalizes over uncertain assignments [21], and PHD/CPHD filters model an unknown number of targets using random-finite-set Bayes filtering [38, 55, 57]. Particle-filter and MCMC variants further support nonlinear dynamics, interacting targets, and variable target counts [29, 41, 56]. These methods are relevant because they preserve multiple explanations under streaming uncertainty, but their hypotheses usually represent physical targets or target sets. In contrast, *IncidentLens* represents hypotheses as latent incident explanations: an incident type, source region, start time, propagation behavior, and supporting or contradicting multimodal evidence. *IncidentLens* also builds on multisensor fusion and spatiotemporal Bayesian filtering. General fusion frameworks study how to align, associate, and combine uncertain observations from multiple sources [24], while belief-function methods provide alternative mechanisms for combining partial evidence [15]. Spatiotemporal filters have been applied to urban and environmental settings such as traffic-volume prediction [42], disaster mobility prediction [52], and remote-sensing image fusion [62]. Unlike these systems, *IncidentLens* does not simply fuse measurements of a known target or field. It uses semantic observations, incident-specific source priors and propagation models, backward proposal of candidate causes, and stability-based clustering to produce explanatory city-scale incident records. Thus, *IncidentLens* is closer to abductive spatiotemporal evidence fusion for urban incidents than to conventional multi-target tracking or single-field sensor fusion.

Complex and compositional event detection. Complex event processing (CEP) systems compose low-level events into higher-level patterns using user-specified rules or queries. Cayuga [14], for example, provides an event language and automata-based execution model for continuous queries over high-throughput streams. Spatiotemporal logic languages [23, 37] add primitives for spatial and temporal selection, aggregation, and ordering. Recent work has combined neural models with CEP [68] or used neurosymbolic architectures to connect neural perception with symbolic event programs [36, 54, 61]. These systems are closest to our goal of connecting perception and symbolic event reasoning, but they generally require predefined event programs or known event patterns. *IncidentLens* differs by first inferring primitive incident records from sparse multimodal evidence and then reasoning over merge, composition, correlation, and co-occurrence relationships among those records.

Smart-city event platforms, event schemas, and foundation models. Smart-city middleware and analytics frameworks have integrated heterogeneous IoT and social streams for event detection and decision support. CityPulse, for example, provides a large-scale data analytics framework for smart-city streams and applications [7]. Such systems motivate the need to combine urban data sources, but they generally emphasize stream integration, querying, or application-level analytics rather than maintaining latent source hypotheses that explain displaced multimodal evidence.

Event-schema induction in NLP provides another view of compositional events. Prior work extracts events and induces schemas from text [25], represents temporal complex event schemas as graphs [31], and uses LLMs to induce open-domain hierarchical event schemas [32]. MultiHiEve [9] extends this idea to video and text, extracting fine-grained event hierarchies across modalities. These works reason about event structure, but primarily from textual or curated multimodal event mentions rather than streaming city sensors with incomplete observability.

Recent work has also explored LLMs and foundation models for urban sensing and management. UrbanGPT [33] and UrbanLLM [27] adapt LLMs to urban spatiotemporal prediction and planning, while ReFound [60] learns multimodal urban region representations and AgentSense [22] uses LLM agents for participatory urban-sensing task assignment. Closer to incident detection, Liao et al. [34] use vision-language models for traffic accident anticipation and localization. These systems demonstrate the promise of foundation models for urban reasoning, but our use of LLMs is more constrained: they map inputs to a fixed ontology or assess bounded structured summaries, while the main incident inference state is maintained by the symbolic hypothesis engine.

7 Conclusion

This paper presented *IncidentLens*, a streaming system for city-scale incident detection from sparse, indirect, and multimodal observations. Rather than treating sensor anomalies or local reports as independent events, *IncidentLens* maintains competing incident explanations over space and time, updates them with semantic observations, and emits stable event-level records.

The main contribution is the combination of persistent source-hypothesis state with a constrained incident-function interface for source priors, propagation, and clustering. This lets different incident types express different physical or semantic assumptions while sharing the same online inference engine. Across synthetic and real Los Angeles incidents, the results show that this structure improves event-level detection and reduces false positives relative to simpler observation-level, clustering, and hotspot baselines.

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A Methodology Details

A.1 Incident Memory and Composition

Incident memory stores recently emitted incident records. Its primary role is not to discover new incidents, but to maintain temporal consistency after incident discovery. In particular, memory prevents duplicate reports when a stable cluster temporarily splits, when nearby hypotheses are reclustered across successive windows, or when new evidence extends an already reported incident. When a new stable record is emitted, *IncidentLens* compares it against records in memory. If the records have compatible incident types, overlapping source or affected regions, overlapping active intervals, and shared or redundant supporting observations, the new record updates the existing memory entry rather than producing a duplicate incident report.

We distinguish three relationships among records. A *duplicate-of* relationship indicates that two records are alternative detections of the same underlying incident; these records may be merged in memory. A *part-of* relationship indicates that one detected incident is evidence for a broader composite incident, such as several local fire records that are later associated with a regional wildfire, or multiple localized public-safety records that are later associated with a larger coordinated incident. A *co-occurring-with* relationship indicates that records are close in space or time but lack enough evidence to justify merging or composition.

Only duplicate handling is part of the core detection path. Part-of links are created conservatively after event-level incident records already exist, and only when there is explicit evidence supporting the relationship, such as shared external reports, compatible incident schemas, overlapping response context, or persistent spatial-temporal coupling across records. These links do not create the initial incident records and do not increase the detection score of their constituent incidents in the main evaluation. Composition is therefore a secondary post-processing capability: *IncidentLens* first infers event-level incident records from observable effects, and only then considers whether some records should be linked into a larger composite incident.

A.2 Semantic Effect Ontology

Table 9 gives representative entries from the semantic effect ontology used by the observation extractor. The ontology is fixed before evaluation. Extractors assign strengths only to known effects, while incident inference is performed by the hypothesis engine.

Table 9: Examples from the semantic effect ontology. Extractors assign strengths only to known effects; incident inference is performed by the hypothesis engine.

Effect	Modalities	Example extraction
smoke/haze	camera, PM2.5, text	Visual haze label, PM2.5 threshold, report mention
fire/flame	camera, text	Visual flame label or explicit report mention
congestion	traffic, camera, text	Speed drop, occupancy increase, stopped vehicles, report mention
water/flooding	camera, weather, text	Visual water, rainfall context, flood report
crowding	camera, text	Visual crowd label or public-report mention
road blockage	traffic, camera, text	Lane occupancy, stopped traffic, road-closure report

A.3 Incident Specifications

Table 10 shows example specifications for a few different types of incidents.

B Synthetic Data Construction

We use a synthetic multimodal incident simulator to generate controlled Los Angeles incident scenarios with known ground truth. The simulator is designed to produce observations that resemble heterogeneous urban sensing data while retaining exact knowledge of the underlying incident source, affected region, time interval, and observation provenance. This allows us to evaluate behaviors that are difficult to measure reliably in real data, such as whether a method can infer an incident source from displaced indirect effects, handle missing modalities, or remain robust when only a subset of sensors observes the event.

Incident structure. Each simulator run begins from an incident request, such as *wildfire*, *urban fire*, or *large civil protest*. In the singular setting, the request corresponds to one localized incident. In the composite setting, a high-level request is decomposed into two related localized sub-incidents. For example, a broad wildfire scenario may contain spatially separated fire or smoke impacts, while an urban disturbance scenario may contain both a public-safety incident and a traffic disruption. Each localized incident is assigned an incident type, a seed location, a short scenario description, and a temporal progression.

Spatial grounding. Each localized incident is bound to a Los Angeles-area geographic region rather than treated as an abstract text prompt. The simulator resolves the seed location into a polygonal region and uses that region as the spatial support for ground truth. If a resolved region is too small to support meaningful sensor placement, the simulator expands it around its representative point so that sensors can be placed over a nontrivial area. This produces incident cases with explicit source regions and affected regions, allowing evaluation to compare predicted spatial support against known ground truth.

Sensor placement. The simulator places physical sensors relative to the incident region. Sensors inside the region are used to generate observations that can directly reflect incident effects. When applicable, the simulator also places outside-region sensors as negative controls. These outside sensors use modalities that would be relevant to the incident if they were nearby, but because they are placed beyond the incident boundary, they report normal or background conditions. This design prevents the dataset from containing only positive evidence and tests whether a detector can distinguish incident-related observations from unrelated nearby activity.

Sensor layouts are fixed across the steps of a simulator run. Thus, repeated observations from the same source correspond to stable physical sensors over time rather than newly invented locations at each step. This is important for evaluating temporal consistency, sensor provenance, and source-level evidence aggregation.

Image observations. For visual observations, the simulator uses real camera imagery as a substrate and edits it to reflect the synthetic incident. Camera imagery is drawn from CCTV or ALERTCalifornia sources depending on the incident context. Wildfire-like incidents preferentially use ALERTCalifornia when available, while most urban incidents use CCTV. The simulator then generates edited images that reflect the incident state at the current step. In our batch generation, image editing is capped at two edited camera frames per simulator step.

The simulator also assigns each camera observation a plausible sensor pose. Camera locations are sampled relative to the incident region, and the camera direction is oriented toward the incident area. This means that image observations are not merely ungrounded pictures; they are associated with a sensor location, viewing direction, source type, timestamp, and incident relation.

Time-series observations. The simulator generates structured time-series observations for modalities such as air quality, weather, and traffic. Air-quality observations can reflect smoke, chemical plumes, or other atmospheric effects. Traffic observations include measurements such as average speed and occupancy, allowing incidents to produce congestion, road disruption, or abnormal mobility patterns. Weather observations provide contextual environmental conditions when relevant.

Inside-region sensors are allowed to express the incident effect. Outside-region sensors are constrained to report background values, such as normal air quality or ordinary traffic speeds. This produces both positive and negative evidence within the same simulated environment and helps test whether the detector relies on spatially and temporally coherent evidence rather than individual anomalous readings alone.

Textual observations. The simulator also generates public-report-like textual observations, including news-style reports, citizen reports, and social-media-like posts. These reports are

Table 10: Example incident-function specifications. These encode compact incident-specific assumptions, not separate end-to-end detectors.

Incident	Effects	Source priors Q_c	Propagation models P_c	Clustering D_c
Wildfire	fire, smoke, PM2.5, road closure	vegetation, terrain, fire risk, sensor coverage	fire: local/field decay; smoke/PM2.5: anisotropic field with wind when available; road closure: network-style	broad spatial support; affectedness overlap
Road accident	congestion, stopped traffic, emergency activity, road closure	roads, intersections, highways, traffic baselines	congestion: downstream network model; emergency activity/blockage: local model	tight source distance; road topology
Flood	water, road closure, congestion, public reports	low elevation, flood zones, drainage, roads	water: field/static terrain model; congestion/closure: network-style	affectedness overlap; terrain consistency
Crowd disruption	crowding, slowed traffic, alerts	venues, transit hubs, public spaces, roads	crowding: local/field model; traffic: network-style	temporal overlap; evidence overlap

conditioned on the current incident state, location, and simulation step. Textual observations may describe direct incident effects, public response, road closures, evacuations, or other secondary consequences. News-style reports may appear later in an incident timeline, while citizen and social reports can appear during intermediate stages. This creates a mix of early, noisy, partial reports and later, more explicit descriptions.

Temporal progression and emitted observations. Each incident evolves over a sequence of simulation steps. At each step, the simulator determines the current incident state, active sources, and relevant modalities, then dispatches modality-specific generators for images, time series, and text. Each emitted observation is written to a common observation log with its incident identifier, step, timestamp, modality, source, sensor metadata, spatial role, and file path when applicable. Ground-truth metadata is saved separately, including the incident region, source region, and scenario state.

Separation from the detector. The simulator and detector use separated scenario specifications. The detector does not receive the simulator prompt, hidden incident state, ground-truth region, or generation labels. It receives only the emitted observations and associated sensor metadata, matching the interface used for real data. This separation reduces circularity: the synthetic dataset provides complete ground truth for evaluation, but the detection system must infer incidents from observations rather than from simulator internals.

Overall, the simulator produces a controlled multimodal benchmark in which every observation is tied to a known incident, time, source, and spatial region, while still preserving realistic heterogeneity across cameras, structured sensors, and public textual reports.

B.1 Implementation Details

For the observation models, we use GPT-4.1-mini to convert raw sensory inputs into structured observations used by downstream components. For anomaly detection, we use modality-specific models: all-MiniLM-L6-v2 from Sentence-Transformers for text embeddings, OpenAI CLIP for image embeddings, and the Python River package for online time-series anomaly detection. These anomaly detection models are run on a GPU workstation equipped with 2× NVIDIA RTX A5000 GPUs and an Intel Core i7-5930K CPU running at 3.50 GHz. For simulation, we generate image observations using Gemini 3.1 Flash Image Preview, and generate text and time-series modalities using GPT-4.

C Dataset Information

Tables 12 summarizes the concrete data sources and dataset statistics used in our real dataset. The dataset has intermittent missing intervals caused by operational disruptions, API changes, and incomplete upstream data. Some modalities also became available only after the start of the study period, resulting in unequal temporal coverage across sources. These limitations make the real-data evaluation imperfect, but they also reflect realistic heterogeneous sensing conditions under partial observability.

D Labelling for Real Data

We process news articles through a staged filtering and consolidation pipeline, with human annotators at the final step. First, we scan GDELT/GKG records and retain only article URLs whose metadata contains relevant emergency-event themes and whose location annotations predominantly refer to Los Angeles. We additionally remove obvious off-topic URL categories and deduplicate articles using normalized URL slugs.

Table 11: Synthetic Los Angeles incident dataset statistics. Singular cases contain one incident. Composite cases were generated with two localized sub-incidents per case. Image generation is capped at two edited camera frames per simulator step.

Split	Incident structure	Incident classes / distribution	Cases	Generated modalities	Total data
Synthetic singular	One incident per case	8 classes: earthquake damage, flood, hazardous material release / chemical plume, large civil protest, severe storm damage, terrorist attack, urban fire, wildfire. Approximately 24 cases per class, with extras.	197	Camera images from CCTV / ALERTCalifornia with synthetic edits; time-series data for air quality, weather, and traffic speed / occupancy; text reports from news, citizen, and social sources.	3.5 GB
Synthetic composite	High-level case with two sub-incidents	4 high-level prompt classes: civil protest (30), urban fire (30), terrorist incident / attack (7), wildfire (30).	97	Same modalities as singular cases. Each composite case contains two spatially separated sub-incidents generated under a shared high-level scenario.	3.6 GB
Total synthetic	Singular + composite	8 singular classes and 4 high-level composite prompt classes.	294	Multimodal synthetic observations spanning edited camera imagery, structured time series, and textual public reports.	7.1 GB

Notes. "Cases" denotes the number of simulator runs. Camera observations are produced from real CCTV or ALERTCalifornia source images using synthetic image edits. The simulator uses at most two edited camera frames per step. Time-series sensors are placed inside the incident region, with additional outside-region negative-control sensors when applicable.

Table 12: Concrete sensing and information sources used in the real-world Los Angeles dataset.

Source	Modality	Sensors / sources	Avg. reports/day	Total data
Caltrans CCTV [2]	camera	296 cameras	28,416	~260 GB
ALERTCalifornia [1]	camera	39 cameras	1,872	~100 GB
PurpleAir [6]	air quality	50 sensors	1,200	~30 MB
OpenWeatherMap [4]	weather	9 locations	216	~360 MB
Caltrans PeMS [10]	traffic	4,888 stations	1,407,744	~6 GB
CitizenApp [3]	public reports	variable	~145	~9 MB
X [59]	social / public safety	6 accounts	~225	~17 MB
GDELT / news [5]	news metadata	variable	~11,100	~6.5 GB

For each retained article, we fetch the webpage, remove boilerplate HTML, and extract likely article text from article-body containers, falling back to the page body when necessary. Articles with insufficient extracted text are discarded. When direct HTTP fetching fails because of blocking or timeout errors, a Selenium-based browser fallback is used.

The extracted article text is then passed to a structured LLM parser that returns candidate incident events. Each extracted event must have an allowed incident type, a specific human-readable location, and an identifiable start time. Events without a valid type, location, or start time are discarded. We further remove events whose inferred start date is implausibly far from the article publication date.

The remaining events are geocoded and clustered into candidate incidents using spatiotemporal proximity. Events are merged when they occur close in time and space, and each cluster stores its source articles, event metadata, representative location, canonical start time, and provenance. We then apply a sensor-observability filter: for each incident type, we identify relevant sensing modalities and test whether corresponding data are available on the incident date and/or spatially intersect the incident region.

Finally, we clean and deduplicate the incident labels. This step rejects generic placeholders, legal/procedural coverage, aftermath-only stories, hypothetical threats, and records whose names or types contradict the allowed incident categories. Aggregate records, such as broad wildfire complexes or protest waves, are separated from concrete low-level incidents. Duplicate low-level incidents are merged conservatively using type, location, date, URL, and named-incident evidence, with safeguards against merging distinct named fires or protests at different locations. The output records include the cleaned incident name, type, location, start/end time, supporting article evidence, and earliest article publication time. This produces a reasonable number of incidents which are then annotated by humans. Our authors researched each describe incident to perform a final level of de-duplication and filtering, and correction for spatial and temporal labels.

E Real-Data Incident Matrix

The real-data labels are weak external incident records, so per-incident F_2 is less interpretable than the aggregate score: each row contains one ground-truth label, and the false-positive term reflects incident-local same-type clutter rather than the global false-positive count. We therefore replace the incident-specific F_2 bar plot with a detection-status matrix that directly shows whether each method recovered each named real label and whether the matched prediction appeared before the earliest external article timestamp.

Table 13 uses the same configured low-real matching policy as the aggregate real-data table. In this run, all 29 weak labels had cached geometry and the evaluator's optional spatial gate was enabled; the table should therefore be interpreted as incident-level recovery under the actual reported matching protocol, not as a separate non-spatial benchmark. Late fusion and text-only clustering are omitted because they detect zero real labels in this run.

The matrix highlights two patterns that are partially hidden by aggregate metrics. First, *IncidentLens* detects the broadest set of named labels, including 10 pre-article detections, while direct observation detects 7 labels and 6 pre-article detections. Second, the density-based baselines mostly detect only the highly observed Palisades Fire and the Waymo urban-fire label; SaTScan and Hawkes each recover only one real label. This supports the aggregate conclusion that local density and temporal-clustering baselines struggle when observations are sparse, indirect, or temporally fragmented.

F Synthetic Observability Coverage

The synthetic low-level evaluation also computes incident-level observability coverage from a non-circular simulator source inventory. We generated `synthetic_coverage_sources.jsonl` from the normalized simulator reports and disabled prediction-support fallback in the evaluator. The resulting coverage run uses 42,867 source records, covers 181 low-level ground-truth incidents, and uses no incident-type default fallbacks.

Synthetic coverage spans a wider range than the real-data labels: scores range from 0.000 to 0.757, with mean 0.485 and median 0.620. Under the current binning, 44 incidents fall in the low-coverage bin, 94 in the medium-coverage bin, and 43 in the high-coverage bin. We therefore treat the synthetic coverage analysis as a diagnostic for whether a method succeeds only when the emitted source inventory makes an incident easy to observe.

Table 14 shows two useful patterns. First, most simple baselines are strongly coverage-dependent: their low-coverage recall is zero, and their match indicators have high positive correlation with coverage. Second, *IncidentLens* is much less coverage-dependent. Its ordinary recall is the highest overall, it detects 63.6% of low-coverage incidents, and its coverage-match correlation is near zero. The coverage-weighted recall of several baselines rises because those methods primarily succeed on high-coverage cases; therefore, we interpret C-rec. as an observability diagnostic rather than as a replacement for ordinary recall or F_2.

F.1 Coverage Smoke Test

We run a small controlled synthetic smoke test to check that the observability score does not manufacture coverage or performance when no source evidence is available. We replay the same ten synthetic flood incidents under four source-availability settings: no emitted reports, low availability, medium availability, and full availability. Coverage is computed only from the emitted normalized REPORT inventory for each condition, and type-default coverage fallbacks are disabled. The no-coverage condition therefore provides an explicit negative control: it should receive a coverage score of zero and produce no successful detections.

Table 15 validates the intended negative-control behavior. When no reports are emitted, the source inventory is empty, the mean coverage score is exactly zero, and *IncidentLens* produces no predictions, yielding zero recall and zero F_2. Increasing source availability from low to medium substantially improves recall and F_2. The medium and high settings have the same mean coverage score because the noisy-or source and modality components saturate once enough relevant flood evidence is present; the full-availability condition emits more predictions and therefore slightly more false positives. This smoke test is not intended as a cross-type benchmark, since the selected incidents are all floods, but it confirms that the coverage pipeline does not assign observability or performance in the absence of evidence.

G Real-Data Observability Coverage

The real-data evaluation also computes an incident-level observability coverage score $C(e)$ to estimate how much evidence could plausibly be available for each weak label. This score is computed from the real-emitter source inventory rather than from method predictions. In the run reported in Section 4.8, coverage uses 101,668 source records from `real_emitter_coverage_` disables prediction-support fallback, and uses no incident-type default fallbacks. Thus, the coverage score is non-circular with respect to the evaluated detectors.

The 29 real labels have moderate-to-high coverage: scores range from 0.518 to 0.749, with mean 0.617 and median 0.610. There are no low-coverage labels under the current binning; 23 labels fall in the medium-coverage bin and 6 in the high-coverage bin. Since the range is narrow and the sample size is small, we interpret coverage correlations as diagnostic rather than as a calibrated causal model of detectability.

Table 16 shows that coverage weighting does not materially change *IncidentLens*'s recall, while direct observation improves under coverage weighting and declines under inverse-coverage weighting. This suggests that direct observation is more concentrated on high-coverage labels, whereas *IncidentLens* is less tied to the coverage score.

Table 17 gives the clearest interpretation of coverage in this run. Direct observation and density-style baselines are much more successful in the high-coverage bin, while *IncidentLens* detects a comparable fraction of medium-coverage labels. This supports the interpretation that source hypothesis maintenance helps recover incidents when observations are sparse or indirect, rather than only improving cases with dense direct coverage.

H Real Top-Level Incident Support

The real top-level file contains broad incident families rather than complete high-level relation graphs. We therefore evaluate real composition as *top-level support*: a method supports a top-level family if it detects enough of the listed low-level child incidents. In the reported run, the two top-level families are the Los Angeles wildfires and the Los Angeles anti-ICE

Incident label	Type	$C(e)$	*IncidentLens*	Direct	ST	Hotspot	SaTScan	Hawkes
Eaton Fire	Wildfire	0.749	–	○	–	–	–	–
Hurst Fire	Wildfire	0.749	–	–	–	–	–	–
Kenneth Fire	Wildfire	0.749	–	–	–	–	–	–
Lidia Fire	Wildfire	0.749	–	–	–	–	–	–
Palisades Fire	Wildfire	0.749	•	•	○	○	○	○
Waymo driverless taxis burned during anti-ICE protests in downtown Los Angeles	Urban fire	0.666	○	•	○	○	–	–
Anti-ICE protest outside City Hall on Spring Street, downtown Los Angeles	Protest	0.610	–	•	–	–	–	–
Compton anti-ICE protest arson	Protest	0.610	–	–	–	–	–	–
Downtown Los Angeles anti-ICE student walkouts and demonstrations	Protest	0.610	–	–	–	–	–	–
ICE Out Everywhere protests at the Metropolitan Detention Center in downtown Los Angeles	Protest	0.610	–	–	–	–	–	–
Los Angeles freeway protest blockade on the 101 Freeway	Protest	0.610	•	•	–	–	–	–
MacArthur Park immigration raid protest	Protest	0.610	•	–	–	–	–	–
National Shutdown protest in downtown Los Angeles	Protest	0.610	•	–	–	–	–	–
Occupy ICE LA	Protest	0.610	•	•	–	–	–	–
Paramount Boulevard immigration protest clash	Protest	0.610	–	–	–	–	–	–
Reclaim Our Streets rally outside Crenshaw Dairy Mart	Protest	0.610	–	–	–	–	–	–
Whittier hotel protest	Protest	0.610	–	–	–	–	–	–
A Day Without Immigrants protest in downtown Los Angeles	Protest	0.518	•	–	–	–	–	–
Hands Off! protest in Pasadena	Protest	0.518	–	–	–	–	–	–
Hands Off! protest in downtown Los Angeles	Protest	0.518	–	–	–	–	–	–
Tesla Takedown rally in West Los Angeles	Protest	0.518	•	–	–	–	–	–

Table 13: Incident-level real-data detection matrix. • indicates a matched prediction before the earliest external article timestamp, ○ indicates a matched prediction after that timestamp, and – indicates no matched prediction. $C(e)$ is the non-circular real-data coverage score computed from the source inventory. This matrix is more diagnostic than per-incident F_2 because it separates label recovery and timeliness from the incident-local clutter term used by the event-specific F_2 calculation.

Method	Rec.	C-rec.	Low rec.	Med. rec.	High rec.	Spearman $\rho(C, \text{match})$
IncidentLens	**0.762**	0.808	**0.636**	0.723	0.977	0.071
Direct obs.	0.630	**0.845**	0.000	0.755	**1.000**	0.821
ST clustering	0.619	0.831	0.000	**0.734**	**1.000**	0.801
Hotspot scan	0.569	0.766	0.000	0.638	**1.000**	0.716
Late fusion	0.564	0.759	0.000	0.628	**1.000**	0.707
Text only	0.519	0.697	0.000	0.681	0.698	0.594
Hawkes detector	0.381	0.519	0.000	0.340	0.860	0.524
SaTScan bg.	0.370	0.500	0.000	0.426	0.628	0.403

Table 14: Synthetic coverage diagnostics computed from per-ground-truth incident rows. C-rec. weights each incident by its observability coverage $C(e)$. The synthetic inventory contains 44 low-, 94 medium-, and 43 high-coverage incidents. The final column is the Spearman correlation between coverage and whether the method matched the incident.

Condition	Sources	Mean C	Pred.	TP	FP	Recall	F_2
None	0	0.000	0	0	0	0.00	0.000
Low	90	0.591	5	5	0	0.50	0.556
Medium	314	0.609	11	10	1	1.00	0.980
High	807	0.609	14	10	4	1.00	0.926

Table 15: Synthetic coverage smoke test over ten replayed flood incidents. Coverage is computed from the emitted source inventory, not from prediction support. Type-default coverage fallbacks are disabled.

Method	Recall	C-rec.	Inv.-C rec.	Spearman $\rho(C, \text{match})$
IncidentLens	**0.379**	**0.376**	**0.382**	-0.019
Direct obs.	0.241	0.257	0.225	0.359
ST clustering	0.069	0.079	0.060	0.385
Hotspot scan	0.069	0.079	0.060	0.385
SaTScan bg.	0.034	0.042	0.028	0.306
Hawkes detector	0.034	0.042	0.028	0.306
Late fusion	0.000	0.000	0.000	–
Text only	0.000	0.000	0.000	–

Table 16: Coverage-normalized real-data recall diagnostics. C-rec. weights each label by its observability coverage $C(e)$. Inv.-C rec. weights labels by $1/C(e)$ as a sensitivity check that gives slightly more weight to harder-to-observe incidents. The final column reports the Spearman correlation between coverage and whether the method matched each label.

protests. Across these families, 25 listed child incidents are evaluable against the low-real summary.

A top-level family is counted as supported when a method detects at least two child incidents or at least 20% of the evaluable children with at least two children detected. This weak metric is intentionally different from the synthetic composition task: it measures whether the low-level evidence is enough

Method	Medium coverage recall	High coverage recall
IncidentLens	**0.391**	0.333
Direct obs.	0.174	**0.500**
ST clustering	0.000	0.333
Hotspot scan	0.000	0.333
SaTScan bg.	0.000	0.167
Hawkes detector	0.000	0.167
Late fusion	0.000	0.000
Text only	0.000	0.000

Table 17: Recall by observability-coverage bin. The real-data labels contain 23 medium-coverage incidents and 6 high-coverage incidents; no label falls in the low-coverage bin.

Method	Child hits	Child rec.	Top support	Pre-report top support
IncidentLens	**8/25**	**0.320**	1/2	1/2
Direct obs.	5/25	0.200	**2/2**	1/2
Hawkes detector	1/25	0.040	0/2	0/2
Hotspot scan	1/25	0.040	0/2	0/2
SaTScan bg.	1/25	0.040	0/2	0/2
ST clustering	1/25	0.040	0/2	0/2
Late fusion	0/25	0.000	0/2	0/2
Text only	0/25	0.000	0/2	0/2

Table 18: Weak real top-level support from detected low-level child incidents. Top support counts the two broad real top-level families whose listed low-level children are sufficiently recovered.

to support a broad real incident family, not whether the system recovered a complete high-level relation graph.

Table 18 shows that *IncidentLens* recovers the most constituent children overall, including 7 of the 20 evaluable anti-ICE protest children and 1 of the 5 wildfire children. Direct observation supports both top-level families under the weak support rule, but this should be interpreted together with the much larger real-data false-positive count in Table 7. The background scan and clustering baselines recover at most one listed top-level child and therefore do not support either top-level family.

We also ran the two offline composition linkers on merged real low-level prediction rows. When applied to direct-observation outputs, the proximity linker recovers the same 5 child incidents and supports both top-level families, while the same-type temporal-overlap linker recovers 3 children and supports one family. When applied to *IncidentLens* outputs, both linkers emit composite groups but none group enough matched child incidents from a top-level family under this weak metric. We therefore treat the real linker results as an appendix diagnostic rather than as the main real-data composition claim.

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